

Education

University of Santiago de Cali

(Currently studying) Master's in Systems

I am currently a student in the Master of Computer Systems program at the University of Santiago de Cali. My expected graduation date is July 20, 2026.

My research focus is on the development of automated systems for small and medium-sized enterprises (SMEs). My goal is to apply my knowledge and skills to develop systems that automate manual processes and integrate with existing processes within the company.

Systems Engineering - License 171122-0620819 VLL

I graduated from the University of Santiago de Cali in November 2021. During my studies, I participated in robotics and programming events sponsored by the university. In these events, I worked on low-level programming of robotic components and programming challenges. My work served as the basis for the development of new projects, such as machine learning projects for the solution of differential equations and computer vision for the biometric identification of personnel for entry into private areas.

Thesis

My bachelor's thesis, titled "Analysis of the benefits and risks of using code coverage", received the highest possible grade (5.0/5.0) from three judges, my thesis advisor, and the dean of the university. The thesis was registered with the university library.

Experience

ArtemisaTech

We developed an application for the registration and management of community and intercollegiate games for the ministry of sports. The application was developed with React for the user interface, AWS (Functions, API Gateway and Streams) for the services and DynamoDB for the database.



Artemisa Tech

A PSE gateway was implemented for the payment of subscriptions of an existing [PHP/Javascript] application in legacy state. The development was done from scratch using Typescript/NestJs to handle the logic and communication with the PSE services. The implementation was done according to the recommended PSE manual and the service successfully met PSE certification. The service is currently in production use.

A module for invoicing, printing and issuance of electronic invoices was developed for Enel Colombia. React-PDF/JS/TS was used to print invoices, which allows the generation of invoices with customer information, payment, installments, bar code and legal disclaimer. The invoicing module also stores information from Codensa cards and Colpatría portfolio purchases. The implementation of services was done in Java/Kotlin, user interface in React and NoSQL database (MongoDB).

A mobile and web application (Lotto/Sports) was developed for users in Puerto Rico and the United States. The application allows users to buy and bet on lotteries and sports results. The development was done with React Native/Expo and the distribution was done through the App Store (iOS) and Play Store (Android).



Artemisa Tech

A mobile application (Powwi/Banco W) was developed targeting users with banking services. The app allows users to make withdrawals at ATMs and banking correspondents, pay for utilities, buy pings for streaming services and send money to other app users. Development was done with React Native/Expo and distribution was done through the App Store (iOS) and Play Store (Android).

Services were developed for the public transport inspection and control module, which are used to send compliance/non-compliance reports, visit correction and notification, 472 correspondence for business notification of documentation and administrative acts, appointment scheduling and visit planning. The implementation of the services was done with C\# using Azure, and the user interface was developed in Angular.

Statement

I graduated as a Software Engineer from the University of Santiago de Cali in 2021. I am currently pursuing a Master's degree in Computer Systems.

In my professional experience, I have worked in the development of financial systems, both in the public and private sectors. In my previous position at Taylor & Johnson, I worked as a Support, Maintenance, and Development Engineer, where I worked with traditional technologies such as Cobol and Mainframes AS/400 (IBM).

Since 2022, I have been working as a Software Developer at Artemisa, where I have developed mobile, web, and back-end applications using current technologies such as AWS, Azure, React, and NoSQL databases. I have also worked with agile methodologies such as Scrum and XP at this company.

I am a professional with a strong commitment to excellence, innovation, and continuous improvement. I have leadership, teamwork, and problem-solving skills. I am always willing to take on new challenges and learn new things.

Languages, Databases & Framework

C++	Standard 17, 14, 11 and 98	MySQL	(Relational) V8 y 5.7
C#	.Net, .Net Core	MongoDB	(No Relational) V4.4.3
Java	JDK 21, 17, 11 and 8	React Native	
Javascript	ECMA 2022,	AWS/Azure	
Python	Version 3.7		



Highly specialized skills

Finding and repairing memory leaks (Memory Leak) with Valgrind.

Design, improvement and writing of documentation, even if non-existent.

Support in the technical analysis and design of Software based solutions.

Development and execution of projects using MongoDB (NoSQL) as data-base.

Development and implementation of Login systems following the RFC5424 standard.

Development and implementation of test cases (unit tests, integration tests and system tests).

Code coverage reporting and analysis (based on both test cases and binary execution).

Code behavior analysis from binaries (Buffer overflows and debugging to unallowed memory accesses).

Design and implementation of meta-programming for space and runtime efficient binary generation.

Reports and analysis of assembly code prone to improvements (necessary to reduce binary size and improve compile time).

Implementation and evolution of continuous integration platform.
Design and implementation of component and microservices based software architectures.

Simplification and refactoring of large code bases. (Specialized in C/C++ and Java).

Evaluation and testing of new solutions and computing platforms in the





Projects

Development and implementation of PNG image decoder and encoder according to the specifications (PNG Standard v1.2).

Development and implementation of decoder and encoder based on Huffman

Coding algorithm for data binarization and encryption.

Development and implementation of software for conversion of PO files to XML.

Development and implementation of database (CRUD operations support) for working on CVS files.

Development and implementation of A* algorithm (Pathfinding).

Development and implementation of hexadecimal editor for any type of file. (Implemented in Qt Core).

Development and implementation of Web application focused on cell phones.

Development and implementation of two-dimensional graphics engine, using OpenGL and SDL.

Development and implementation of framework for the design, implementation and development of programs with rendering requirements and use of algorithms (Encryption, Compression, Pathfinding, etc ...).



Technological experience

Experience in the development of components and microservices.

Experience in refactoring large code bases.

Experience in refactoring legacy software.

Experience in C/C++ (low level programming). Supporting standard 99, 11, 14, 17 and 20.

Experience in Java 8, 12 and later versions.

Experience with non-relational databases (Mongo DB).

Experience in application development: React, NodeJS, Bootstrap.

Experience web application development: HTML, CSS and Javascript.

Experience in mobile-focused web application development.

Experience with continuous integration environments (Jenkins, Sonar, Mockito, Cucumber, Junit...).

Experience with version control software (Git, Github, Fossy, etc ...).

Experience with version control dynamics (Pull Request, Push, Fork, etc ...).

Experience in web page analysis and performance improvement (Chromium Development Tools and Lighthouse).

Experience in the development of documentation corresponding to the executed part of the project.

Experience in the participation of tasks of putting into production of the



Technological experience

developed systems (preparation of delivery, implementation, support).

Experience in maintenance of information systems.

Experience in the technical documentation of the project.

Customer service skills

6 years of experience in customer service and customer care (both physical and virtual).

3 years in team coordination and team multi-tasking.

Excellent charisma, attitude, interaction and interaction with customers, teams and individuals.

Excellent personal presentation.

Excellent vocabulary and persuasion skills.

Excellent writing skills, both in Spanish and English.

